

Trust-Calibrated Multimodal Industrial Anomaly Intelligence Platform

Confidence-Calibrated Multimodal Fusion for Robust Industrial Anomaly Detection, Root-Cause Reasoning, and Predictive Maintenance Under Missing and Noisy Sensor Modalities

Applied Research + Engineering Project Report

Executive Positioning

This report presents a proposed research and engineering system rather than a completed experimental study. No numerical experimental results are claimed. The project design, architecture, datasets, evaluation plan, technology stack, risks, and two-person division of work are grounded in the supplied project design document and accompanying specification. The source document explicitly separates the research core—trust gating, calibration, and trust recalibration—from the surrounding platform layer of RAG, LLM reasoning, dashboard, and deployment infrastructure.

The novelty claim is deliberately narrow: the work does not claim to invent multimodal fusion, dynamic modality weighting, missing-modality learning, confidence calibration, RAG, or human-in-the-loop learning. Instead, it proposes their integration around industrial anomaly detection, with calibration treated as a first-class optimization/evaluation concern, a controlled degradation benchmark, and persistent asset-specific trust updated from human correction history.

Abstract

Industrial inspection systems increasingly combine visual inspection with machine telemetry, historical inspection records, and technical documentation. The practical difficulty is that these evidence sources are not uniformly reliable: cameras can be blurred or occluded, sensors can be noisy or unavailable, telemetry can become stale, historical evidence can be sparse, and different modalities can contradict one another. A multimodal system that simply averages or concatenates evidence may therefore remain numerically confident even when the information available to it has materially degraded.

This project proposes a Trust-Calibrated Multimodal Industrial Anomaly Intelligence Platform in which each modality is encoded independently, assigned an explicit quality representation, and passed through a learned per-instance reliability gate. For modalities V (vision), S (sensor telemetry), H (history), and D (document evidence), the system estimates trust weights that influence multimodal fusion. The research core further incorporates calibration-aware optimization, using a differentiable proper scoring term such as the Brier score during training and held-out temperature scaling for deployment, while Expected Calibration Error (ECE), Brier score, and reliability diagrams remain central evaluation measures. Missing, noisy, stale, and contradictory modalities are introduced systematically through a controlled degradation benchmark.

A second research component is a persistent trust memory at the machine/plant level. Human corrections are logged together with modality availability, quality, trust weights, predictions, and outcomes. A guarded periodic update mechanism adjusts asset-specific modality priors so that repeated deployment experience can influence future gating decisions. The proposed evaluation compares image-only, static fusion, hybrid/cross-modal baselines, dynamic gating, calibration-aware gating, and the full human-feedback system across 0%, 25%, 50%, and 75% modality dropout, contradiction, noise, and staleness conditions. The intended contribution is a measurable and falsifiable framework for determining whether dynamic trust can improve both robustness and confidence honesty under degraded multimodal evidence.

1. Introduction

Traditional industrial computer-vision inspection often reduces a complex physical process to an image-level decision such as defect/no defect. That formulation is useful for high-throughput screening, but it does not by

itself answer why a defect may have occurred, whether the operating conditions are abnormal, whether the defect is worsening, whether the observation is consistent with prior machine history, or what documented maintenance procedure should follow.

Multimodal industrial intelligence addresses this limitation by combining complementary evidence. Vision can provide spatial and surface information; vibration, temperature, pressure, RPM, current, voltage, and acoustic measurements can describe operating conditions; historical records can expose temporal patterns; and manuals/SOPs can provide procedural context. The supplied design document therefore defines the platform around four data families: camera/video, telemetry, historical records, and technical documents.

2cite2turn0search02turn0search32

The practical challenge is that these modalities do not fail symmetrically. A camera may be partially occluded, a vibration sensor may have a communication gap, telemetry may arrive with delay, a historical record may be sparse, or retrieved documentation may have low relevance. A particularly important case is contradiction: the camera may indicate a crack while vibration remains nominal. A fixed fusion rule cannot explicitly represent the fact that one evidence source has become less trustworthy in the current observation.

This motivates the central research question: Can a multimodal industrial anomaly detection system learn to dynamically discount unreliable evidence, remain calibrated under increasing modality degradation, and improve its modality trust behavior over time using machine-specific human correction feedback?

The project therefore has two simultaneous identities. First, it is an applied research study centered on reliability-aware fusion, calibration, degradation robustness, and adaptive trust. Second, it is an engineering platform that exposes the research outputs through anomaly localization, severity estimation, root-cause ranking, predictive maintenance, RAG-grounded explanation, human review, and a deployable dashboard.

The claim is intentionally not that any individual mechanism is unprecedented. Existing work already demonstrates multimodal industrial anomaly detection, cross-modal feature mapping, missing-modality learning, probabilistic multimodal fusion, dynamic quality-aware fusion, and uncertainty-aware weighting. The proposed research contribution is at the level of the combination, objective, evaluation protocol, and deployment feedback loop.

2. Problem Statement

Let the four principal evidence modalities be V for vision, S for sensor telemetry, H for historical information, and D for technical/document evidence. For an industrial asset a , the ideal system estimates the posterior probability of an anomaly state Y :

$$P(Y | V, S, H, D)$$

The formulation must remain valid when any subset of modalities is unavailable, noisy, stale, corrupted, or contradictory. We therefore associate each modality with a reliability variable $r_i \in [0, 1]$, where $i \in \{V, S, H, D\}$. These variables are not the final prediction confidence. They express the amount of influence that the corresponding evidence source should receive in the fusion process.

$$r_V, r_S, r_H, r_D \in [0, 1]$$

A quality estimator produces observable evidence about modality condition, q_i . A trust gate then maps quality, modality representation, asset context, and a persistent asset-specific trust prior into a gate value g_i . The normalized fusion weight w_i is derived from these gates. A missing modality is not silently imputed in the primary design; its gate is forced to zero and the event is explicitly logged.

$$g_i = \text{Gate}(q_i, z_i, c_a, p_i(a)); \quad w_i = g_i / (\sum_j g_j + \epsilon)$$

$$z_{\text{fused}} = \sum_i w_i z_i$$

Three evaluation concepts must be separated. Prediction accuracy asks whether the model is correct. Robustness asks whether performance degrades gracefully when evidence quality changes. Calibration asks whether a reported probability corresponds to empirical correctness. A model can therefore have acceptable AUROC while being badly overconfident under modality dropout. The proposed study treats that distinction as central rather than incidental.

3. Research Gap

3.1 Static versus adaptive fusion

Multimodal industrial anomaly detection has progressed from simple feature concatenation toward hybrid fusion, cross-modal feature mapping, and architecture-search approaches. M3DM, for example, combines RGB and 3D information through a hybrid feature/decision fusion design rather than naive concatenation. Crossmodal Feature Mapping learns to map one modality's features to another and detects inconsistencies between observed and mapped features. These approaches establish that multimodal representation design matters, but they do not establish the proposed asset-specific trust loop or the proposed calibration-centered degradation protocol. [\[cite?turn0search0?turn0search3?\]](#)

3.2 Accuracy versus calibration

The proposed gap is not that calibration metrics are unknown. Calibration is a mature topic, and temperature scaling is a standard post-hoc technique. The gap is the evaluation question: how does confidence honesty change as industrial evidence is progressively removed or corrupted? The study therefore treats ECE, Brier score, and reliability diagrams as primary outputs alongside accuracy metrics. Temperature scaling is used as a held-out calibration mechanism rather than incorrectly backpropagating standard binned ECE.

3.3 One-time reliability versus persistent trust

Dynamic weighting methods can estimate reliability for an individual observation. The proposed system adds a separate persistent state: a machine/plant-specific prior that summarizes historical evidence about how useful each modality has been for that asset. This is not assumed to replace the learned gate. The global gate learns general behavior; the persistent prior provides deployment-specific context.

3.4 Offline benchmark versus deployment feedback

Standard industrial anomaly benchmarks do not naturally contain long-term operator correction histories. The proposed platform therefore logs review decisions and uses them in a controlled trust-recalibration loop. This makes the project a closed-loop deployment study rather than a single-pass benchmark.

3.5 Unified degradation protocol

The principal experimental axis is 0%, 25%, 50%, and 75% modality dropout, complemented by noise, staleness, and contradiction injection. The purpose is not to manufacture an artificial claim that factories behave exactly like the synthetic benchmark. The purpose is to create a reproducible stress test in which reliability and confidence can be measured under controlled degradation.

4. Related Work

4.1 Industrial visual anomaly detection

MVTec AD remains a foundational benchmark for unsupervised industrial anomaly detection, containing high-resolution images across object and texture categories with pixel-level anomaly annotations. Its design is well suited to establishing a strong vision baseline before adding synthetic telemetry. MVTec AD 2 extends the benchmark to harder inspection conditions, including transparent or overlapping objects, challenging illumination, high normal variation, and very small defects. [\[cite?turn1search7?turn2search6?turn2search8?\]](#)

4.2 Multimodal industrial anomaly detection

Wang et al. introduced M3DM in CVPR 2023 for RGB plus 3D point-cloud industrial anomaly detection. Their hybrid fusion includes patch-wise contrastive feature interaction and a decision-level fusion using multiple memory banks, with point-feature alignment. The work is an appropriate static/hybrid baseline because it demonstrates that fusion architecture materially affects industrial anomaly detection. Its modality setting is RGB + 3D rather than the proposed vision + telemetry + history + documents setting, and its published evaluation is not the proposed calibration-under-degradation protocol. [\[cite?turn0search0?\]](#)

Costanzino et al. proposed Crossmodal Feature Mapping at CVPR 2024. Their method learns to map features from one modality to another using nominal samples and detects anomalies through observed-versus-mapped feature inconsistency. The authors emphasize reduced memory and inference cost relative to memory-heavy multimodal approaches. Importantly for this project, the paper states a limitation: its paradigm is multimodal-only and requires both modalities at training and test time. This makes it a strong conceptual baseline for cross-modal consistency while also highlighting why missing-modality handling requires explicit design. [\[cite?turn0search3?turn0search132?\]](#)

Cheng, Luo, and Zhang's 2025 UCF method combines uni-modal reconstruction and cross-modal mapping to capture intra- and inter-modal relationships in RGB and point-cloud anomaly detection. It reinforces the importance of both modality-specific and cross-modal representations. The proposed project differs primarily in the reliability/control axis: its central mechanism is not a richer RGB–3D representation but an explicit reliability gate coupled to calibration and persistent asset trust. [\[cite?turn1search4?turn1search8?\]](#)

Long et al. provide an architectural study of multimodal fusion for 3D anomaly detection and introduce a neural architecture search approach, 3D-ADNAS, to jointly explore fusion strategies and modality-specific modules. This work is particularly relevant to the architectural dimension of the problem. The proposed project shifts the question from 'which fusion topology is best?' toward 'how should influence change when evidence reliability changes?' [\[cite?turn0search2?\]](#)

The 2025 Information Fusion survey by Lin et al. provides a broader state-of-the-field view of RGB, 3D, and multimodal unsupervised industrial image anomaly detection and explicitly surveys fusion strategies and research challenges. It supports the positioning of the present project as an extension of an already active multimodal anomaly-detection field rather than a new invention of multimodal IAD itself. [\[cite?turn3search3?\]](#)

4.3 Missing-modality learning

CMDIAD is especially close to the proposed missing-modality problem. Sui et al. propose cross-modal distillation for incomplete multimodal industrial anomaly detection, using a multi-modal-training/few-modal-inference pipeline and hallucinating missing modality information. The published study demonstrates that incomplete multimodal IAD is already an active research problem. The proposed system therefore must not claim that missing-modality robustness itself is novel. Its distinction is that the principal question is

calibration and trust behavior under degradation, not only whether accuracy can be preserved by reconstructing missing information. [\[2\]cite\[2\]turn5search0\[2\]turn5search2\[2\]](#)

A 2025 RGBD industrial anomaly detection study likewise reports hierarchical fusion that tolerates missing inputs. This further weakens any novelty claim based simply on 'fusion that survives missing data.' The project should instead evaluate whether confidence remains appropriately conservative as modalities disappear. [\[2\]cite\[2\]turn3search0\[2\]](#)

4.4 Adaptive modality weighting and quality-aware gating

DUAL-Health is an adjacent example of uncertainty-aware multimodal fusion under noisy outdoor health-monitoring conditions. It quantifies modality uncertainty and customizes fusion weights based on uncertainty. The methodological lesson is directly relevant: reliability can vary by modality and condition, and fusion should respond to that variation. The proposed project transfers this principle to industrial evidence while adding an asset-specific persistent trust state and a calibration/degradation evaluation protocol. The adaptive reweighting idea itself is therefore not claimed as novel. [\[2\]cite\[2\]turn2academia71\[2\]turn5academia37\[2\]](#)

Quality-aware optical–SAR fusion provides another adjacent example. QDFNet estimates modality reliability and dynamically adjusts fusion in the presence of degraded or missing modalities. The domain is remote sensing rather than industrial inspection, but the work reinforces the conceptual validity of explicit quality estimation before fusion. Again, the proposed contribution lies in the industrial anomaly setting, calibration-centered evaluation, persistent asset trust, and human feedback loop. [\[2\]cite\[2\]turn4academia48\[2\]](#)

4.5 Trust-aware multimodal fusion

Patel et al.'s Trust-Aware Multimodal Data Fusion for Beirut explosion yield estimation uses a Bayesian fractional-posterior formulation and a Dirichlet prior to learn relative trust across heterogeneous observational modalities. It provides a useful methodological precedent for treating source reliability as an explicit modeling object rather than assuming equal evidence quality. The proposed project differs in domain, persistent machine/plant conditioning, operator-feedback updating, and calibration evaluation under industrial degradation. The work is a 2025 arXiv preprint rather than an established industrial-anomaly benchmark paper. [\[2\]cite\[2\]turn2academia69\[2\]turn2search2\[2\]](#)

4.6 Confidence calibration in machine learning

Guo et al. established the practical importance of neural-network calibration and showed that temperature scaling can be effective as a simple post-hoc method. The present project adopts this established foundation but changes the evaluation context: calibration is tested as a function of modality degradation. Standard ECE is retained as an evaluation metric; a differentiable proper scoring rule such as Brier loss is preferred for the training objective because ordinary binned ECE is not a straightforward differentiable loss.

[\[2\]cite\[2\]turn1search0\[2\]](#)

4.7 Probabilistic ensembling

ProbEn studies multimodal RGB–thermal object detection and derives a probabilistic ensemble from Bayes' rule. It can accommodate missing detections and demonstrates why probability-aware fusion can be preferable to naive averaging. The proposed project is more explicit about learning evidence reliability from quality signals and persistent asset context, and it extends the concern from detector fusion to industrial anomaly confidence. [\[2\]cite\[2\]turn4search2\[2\]](#)

4.8 Adaptive conflict-aware fusion

MAF for attributed-graph anomaly detection uses conflict-aware adaptive fusion and a feedback-refined reweighting loop. Its domain is graph anomaly detection, so it is not a direct industrial-vision precedent. It is

nevertheless relevant evidence that adaptive fusion can be used to resolve heterogeneous and conflicting anomaly signals. The proposed work transfers the general principle into industrial multimodal evidence and adds calibration and asset-specific trust. [2]cite[2]turn4search1[2]

4.9 RAG and LLM-based maintenance reasoning

PARAM demonstrates a broader direction in which RAG and LLM-based agents convert industrial telemetry/anomaly information and maintenance manuals into prescriptive recommendations. In the proposed architecture, this capability remains an engineering and reasoning layer rather than the primary anomaly detector. This separation is intentional: specialized models produce structured evidence; retrieval supplies procedural context; the LLM explains and organizes that evidence. [2]cite[2]turn1academia113[2]

4.10 Research gap and positioning

The literature therefore supports a careful conclusion: the individual mechanisms already exist. Multimodal IAD, cross-modal mapping, incomplete-modality inference, quality-aware gating, probabilistic fusion, trust-aware fusion, calibration, RAG maintenance reasoning, and feedback-driven adaptation are all established or actively developing directions. The proposed research claim is consequently at the combination level: industrial anomaly detection × per-instance trust gating × calibration-aware optimization/evaluation × controlled degradation × persistent asset-specific trust × human correction feedback.

To the best of the present review, the most defensible gap is not 'no one has done this,' but that these elements are not commonly evaluated together around the specific question of confidence honesty under progressively degraded industrial evidence. This claim remains falsifiable: if a future systematic search identifies an equivalent industrial system and protocol, the novelty claim must be narrowed accordingly.

Table 1. Literature comparison

Work	Domain	Modalities	Fusion strategy	Missing data?	Dynamic reliability?	Calibration?	Human feedback?	Persistent asset trust?	Difference
Wang et al., 2023 [1]	Industrial RGB + 3D IAD	RGB, point cloud	Hybrid feature + decision fusion	Not established as primary focus	Fixed/hybrid	Not established	No	No	Use as hybrid/static baseline; proposed work adds trust, calibration and asset memory.
Costanzino et al., 2024 [2]	Industrial RGB + 3D IAD	RGB, point cloud	Crossmodal feature mapping	Limited ; paper states both modalities required at test	No explicit persistent trust	Not established	No	No	Proposed work explicitly handles unavailable evidence and calibrates confidence.
Cheng et al., 2025	Industrial RGB + 3D	RGB, point cloud	Uni-modal reconstr	Not established	Not persistent	Not established	No	No	Focus shifts from representat

[3]	IAD		uction + cross-modal mapping						ion completeness to reliability-aware fusion.
Long et al., 2025 [4]	3D anomaly detection	RGB + 3D	Architecture analysis + NAS	Not primary focus	No persistent trust	Not established	No	No	Proposed work optimizes/conditions fusion by evidence reliability.
Lin et al., 2025 [5]	Survey	RGB, 3D, multimodal	Survey of fusion strategies	Survey	Survey	Survey	No	No	Provides field-level positioning.
Sui et al., 2026 [6]	Incomplete industrial IAD	RGB, point cloud	Cross-modal distillation / hallucination	Yes	Implicit via modality completion	Not central	No	No	Closest missing-modality baseline; proposed axis is calibration honesty.
Hoan et al., 2025 [7]	Industrial RGBD IAD	RGB, depth	Hierarchical multimodal fusion	Yes	Quality/missing robustness	Not established as core	No	No	Shows missing-input robustness is already established.
DUAL-Health [8]	Outdoor health monitoring	Multiple health sensors	Uncertainty-aware dynamic weighting	Addresses noisy/missing recovery	Dynamic	Calibrated uncertainty	No	No	Motivates dynamic reliability; domain differs.
Patel et al. [9]	Explosion yield estimation	Seismic, crater, SAR, VLM images	Bayesian trust-weighted fusion	Robustness to bias	Trust weights	Uncertainty quantification	No	No	Trust precedent; proposed work adds asset-specific feedback and industrial IAD.
ProbEn [10]	Multimodal object	RGB, thermal	Probabilistic ensemble	Yes	No learned gate	Probability-aware	No	No	Motivates probability-aware

	detection		ling						fusion; proposed gate uses explicit quality/trust.
Zhao et al. [11]	Remote sensing object detection	Optical, SAR	Dynamic quality-aware fusion	Yes	Yes	Not central	No	No	Quality gating precedent; different domain and objective.
Gusare et al. [12]	Attributed graph anomaly detection	Structural/contextual/latent	Conflict-aware adaptive fusion	Not industrial missing-modality focus	Yes + feedback	Not central	Feedback loop	No	Supports adaptive conflict handling; domain differs.
Harbola & Purwar [13]	Industrial maintenance reasoning	Telemetry + manuals + web knowledge	RAG + LLM agents	Not fusion-focused	Not central	Not central	Not central	No	Engineering explanation layer, not core detector.

5. Proposed System

The platform is organized into four layers: (1) data layer, (2) multimodal fusion/trust-gating core, (3) reasoning and decision layer, and (4) human-in-the-loop feedback and trust recalibration. This follows the supplied design document and keeps the research contribution visibly separate from the engineering shell.

The primary processing chain is:

Data sources → modality encoders → quality estimation → trust gates → trust-weighted fusion → calibrated prediction → downstream intelligence → RAG/LLM explanation → human review → correction logging → trust recalibration → updated priors → future inference

The four evidence sources are deliberately heterogeneous. Vision and telemetry are continuously sensed, history is accumulated over time, and documents are retrieved conditionally. The fusion core should therefore not treat all four sources as interchangeable feature vectors. Each source has a modality-specific encoder and quality representation before entering the common fusion space.

5.1 Module 1 — Visual inspection

The vision subsystem should support two complementary paths. A supervised object detection/segmentation path handles known defect classes such as scratches, cracks, dents, corrosion, contamination, deformation, or missing/wrong components. An open-set anomaly path handles defects not represented in the labeled taxonomy. The latter is important because exhaustive defect labeling is unrealistic in industrial environments.

5.2 Module 2 — Defect localization

The localization output is a pixel-level anomaly map or segmentation mask rather than a single image-level label. Localization supports human review, severity estimation, and the dashboard's spatial explanation. PatchCore, DRAEM, PaDiM-style methods, or comparable modern anomaly detectors can be used as baselines.

5.3 Module 3 — Quality gates and trust-gated fusion

This is the primary research module. A small gating network estimates per-instance modality influence from quality signals, learned representations, asset context, and the persistent trust prior. A missing modality is explicitly represented as unavailable and receives zero gate weight.

5.4 Module 4 — Severity estimation

Severity is a downstream decision variable, not a synonym for anomaly probability. The proposed tiers are LOW, MEDIUM, HIGH, and CRITICAL. Inputs may include calibrated anomaly probability, localized defect area, operating conditions, machine age, historical frequency, and predicted failure risk.

5.5 Module 5 — Root-cause ranking

Root-cause analysis is explicitly framed as probabilistic or correlational ranking. Candidate causes may include excessive temperature, tool wear, material inconsistency, vibration abnormalities, alignment issues, or process drift. The system must not claim causal proof merely because a variable correlates with an anomaly.

5.6 Module 6 — Predictive maintenance

A telemetry/history model estimates failure probability over a configurable horizon. A seven-day horizon is a plausible demonstration setting, but the API should allow the horizon to be changed. LSTM/GRU or PatchTST-style models are candidates; model selection should be empirical.

5.7 Module 7 — RAG and LLM reasoning

The LLM does not perform the primary anomaly detection. It receives structured outputs from specialized models and retrieved SOP/manual evidence, then produces a human-readable summary, evidence list, and recommended action. The recommendation should cite or identify the supporting document passages. This architecture limits hallucination by constraining the LLM's role to evidence-grounded synthesis.

5.8 Module 8 — Human review and correction logging

Predictions can be routed according to confidence tiers: high-confidence cases may be automatically resolved where policy permits, intermediate cases can require review, and low-confidence or safety-sensitive cases can require manual inspection. Every review event is logged with the complete modality-trust context.

5.9 Module 9 — Trust recalibration

The trust recalibration loop periodically updates the asset-specific modality prior from accumulated corrections. This is intentionally separate from the global gate parameters. The gate learns generalizable behavior; the prior learns local deployment characteristics.

5.10 Module 10 — Dashboard and alerts

The dashboard exposes not only the prediction but also the evidence state: modality availability, quality, trust weight, calibrated confidence, severity, probable causes, failure probability, and supporting SOP evidence.

Making trust visible is essential because otherwise the central research mechanism is invisible to the operator.

Figure 1 — Overall System Architecture

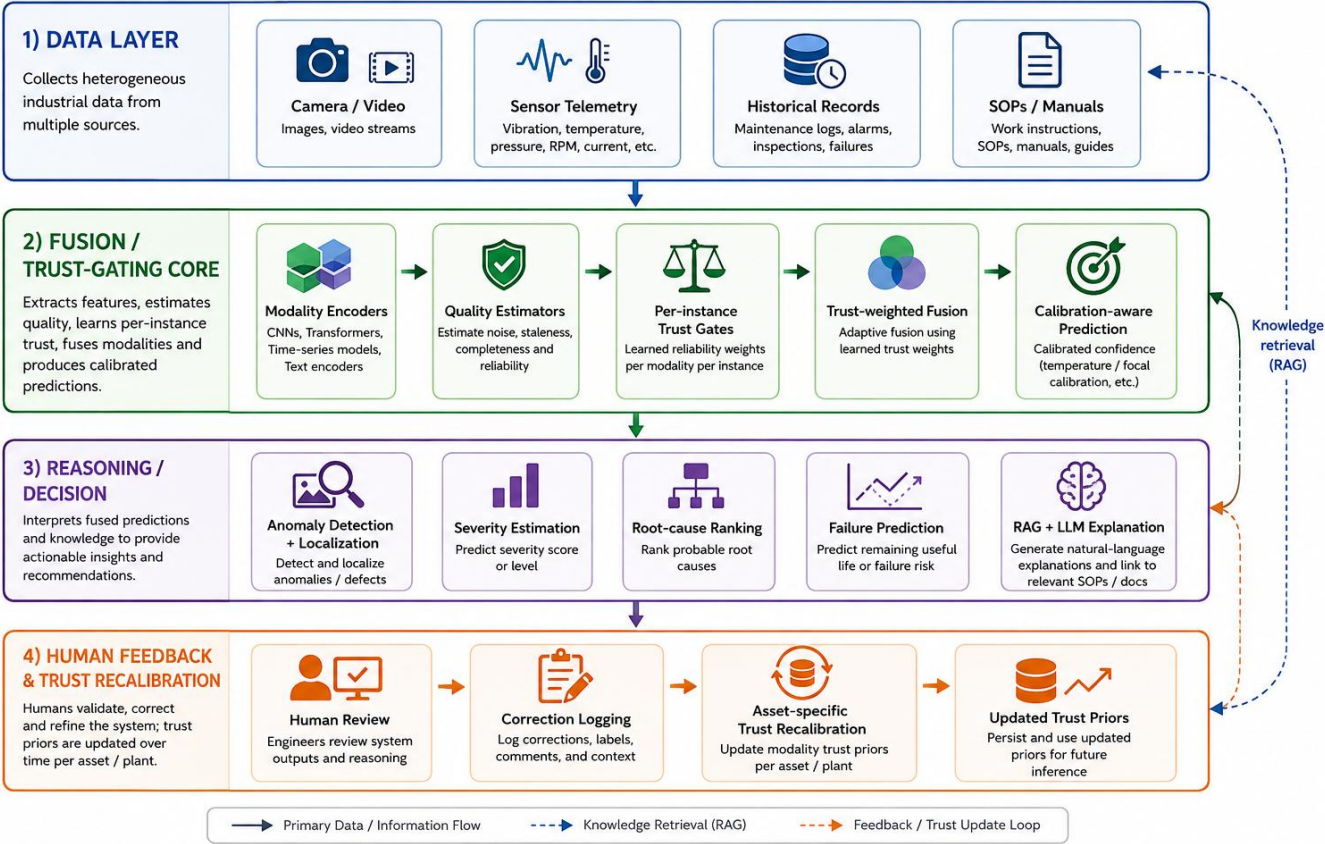


Figure 1. Overall system architecture.

The research core is the trust-gating and calibration path in the middle of the architecture. RAG, LLM, dashboard, and deployment services surround that core and are explicitly treated as engineering/product layers.

6. System Architecture

6.1 High-level architecture

The high-level system consists of four modality inputs, four encoder families, a quality-estimation stage, a trust-gating stage, a fusion and calibration stage, downstream decision modules, and a human-feedback loop. The feedback loop returns to the trust gates rather than directly changing the detector on every human interaction.

Figure 2 — Detailed Trust-Gated Fusion Architecture

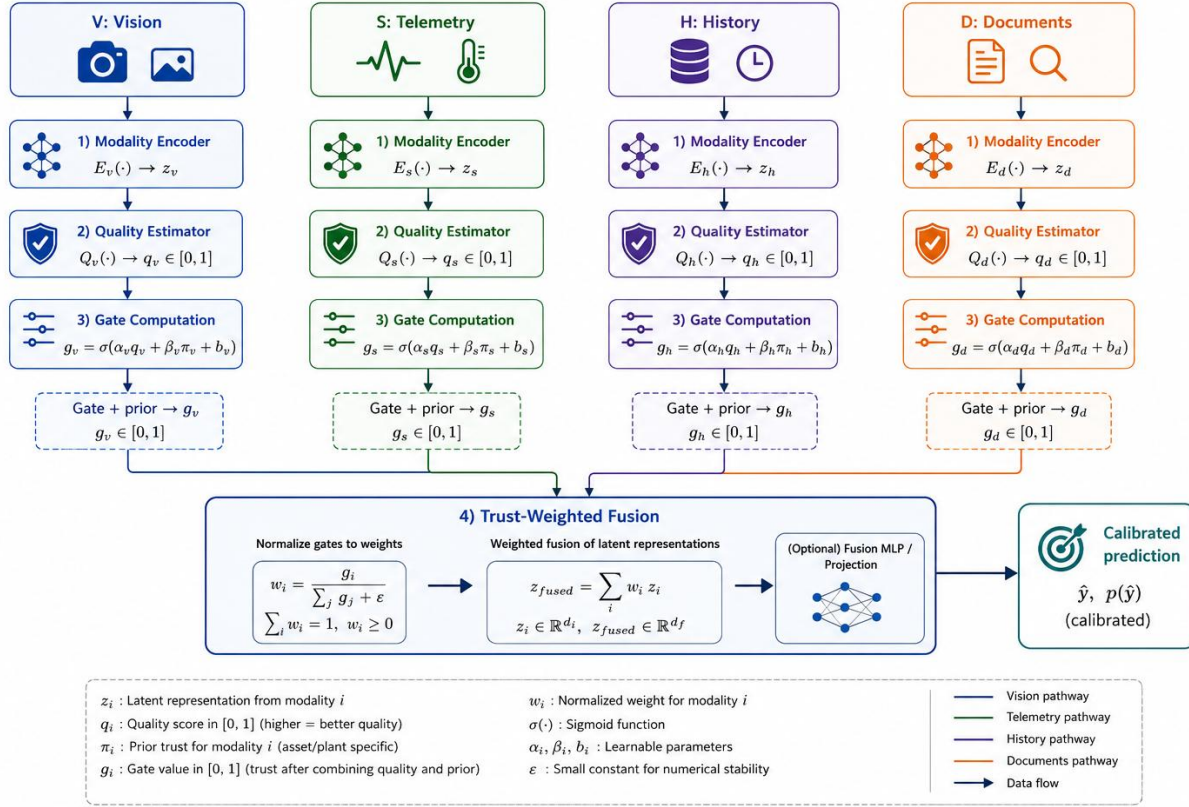


Figure 2. Detailed trust-gated fusion architecture.

Each modality produces an embedding z_i and quality representation q_i . The learned gate combines q_i , z_i , asset context, and the persistent prior $p_i(a)$. The normalized weights then determine the contribution of each modality to the fused representation.

6.2 Research architecture and mathematical formulation

For each modality i , let $z_i \in \mathbb{R}^d$ denote its embedding. Let q_i encode measurable evidence quality. Let $p_i(a)$ denote the persistent trust prior for modality i on asset a . A learned gate can be implemented as:

$$g_i = \sigma(\text{MLP}_i([z_i; q_i; c_a; p_i(a)]))$$

where σ is a sigmoid function and c_a is an asset/context representation. For unavailable modalities, g_i is deterministically set to zero. The remaining weights are normalized:

$$w_i = g_i / (\sum_j g_j + \epsilon)$$

The primary proposed implementation is weighted representation fusion followed by a fusion MLP. Weighted cross-attention is an alternative if experiments show that simple weighted summation loses important cross-modal interactions. The project should choose one primary architecture before the final ablation study so that comparisons remain interpretable.

$$z_{fused} = \text{MLP}_{fuse}(\sum_i w_i z_i)$$

An alternative attention implementation is:

$$z_{fused} = \text{WeightedCrossAttention}(z_V, z_S, z_H, z_D; w)$$

The gate should not be allowed to collapse permanently onto a single modality. Regularization should encourage stable, evidence-dependent weighting and prevent pathological behavior in which one modality receives near-zero influence regardless of its condition.

7. Modality-Specific Quality Estimation

Quality estimation answers a different question from final prediction confidence. Quality asks whether an evidence source is likely to be reliable; prediction confidence asks how likely the final model output is to be correct. A high-confidence prediction can still be wrong if the system is miscalibrated, and a high-quality modality can still legitimately support an anomalous prediction.

Figure 3 — Modality Quality Estimation Pipeline

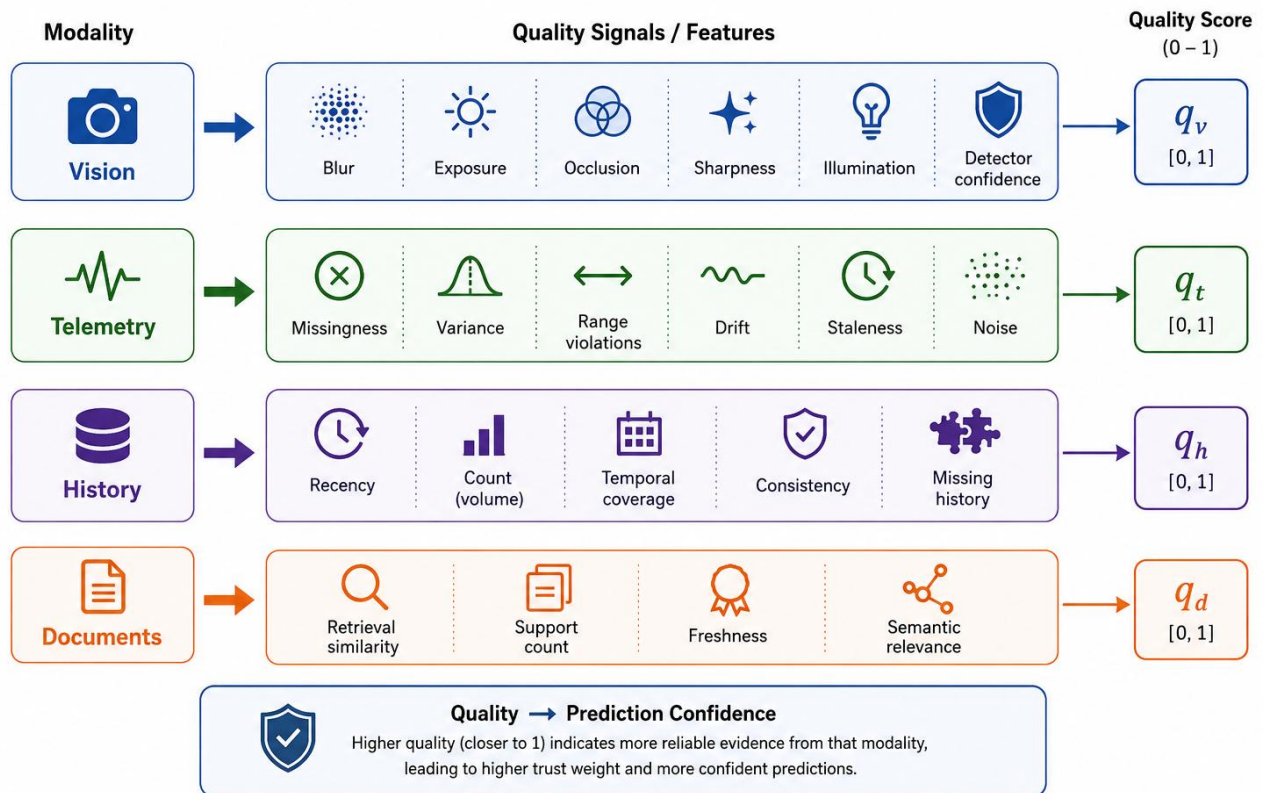


Figure 3. Modality quality estimation pipeline.

Quality indicators are modality-specific. They are transformed into q_i and supplied to the gate; they are not treated as equivalent to the final anomaly probability.

7.1 Vision quality

- Blur/sharpness measures to identify out-of-focus images.
- Exposure and illumination statistics to identify under/overexposure.
- Occlusion estimates for dust, obstruction, or partial visibility.
- Image artifact indicators such as compression or motion artifacts.
- Detector/segmenter confidence as a secondary signal, not a sole quality measure.
- Geometric or alignment checks where multi-camera/3D data are used.

7.2 Telemetry quality

- Missing-sample fraction and communication interruptions.
- Variance and noise statistics over an appropriate rolling window.
- Range violations and physically implausible values.
- Drift relative to historical operating distributions.
- Timestamp staleness and irregular sampling.
- Signal-to-noise or residual-based indicators where reference behavior is available.

7.3 Historical-record quality

- Recency of the latest confirmed inspection.
- Number of observations available for the machine.
- Temporal coverage and gaps.
- Consistency of labels or inspection outcomes.
- Similarity between current operating context and historical context.

7.4 Document/RAG quality

- Retriever similarity score.
- Number of independently supporting passages.
- Document freshness/version.
- Semantic relevance to the current machine/defect context.
- Presence of explicit procedural language in retrieved evidence.

8. Trust Gating Mechanism

Prediction confidence and modality trust are distinct. Prediction confidence asks: 'How confident is the model about its final decision?' Modality trust asks: 'How much should this evidence source influence the decision right now?' The proposed architecture explicitly models both.

Per-instance gating is necessary because the same modality can be useful for one sample and unreliable for another. A clean, sharp camera image may receive high visual trust, while an image with severe glare receives lower trust. A vibration signal with missing samples or timestamp delay may receive reduced trust. A retrieved SOP passage with low semantic relevance should contribute little or be excluded.

The gate should use both local quality and learned representation context. Quality-only gating is insufficient because an apparently clean sensor can still be semantically uninformative for a particular anomaly. Conversely, a noisy modality may still contain complementary information that is useful in combination with another modality. The gate therefore estimates influence rather than enforcing a rigid rule.

- Unavailable modality: force $g_i = 0$ and log the event.
- Noisy modality: reduce influence according to quality and learned context.
- Stale modality: penalize reliability as temporal distance increases.
- Contradictory modality: allow the gate to discount inconsistent evidence and increase uncertainty rather than blindly averaging.
- Persistent asset prior: bias the gate toward historically reliable modality behavior without overriding current quality.
- Stability regularization: discourage abrupt unexplained gate oscillation across near-identical inputs.

$$0 \leq g_i \leq 1$$

$$w_i = g_i / (\sum_j g_j + \epsilon)$$

The denominator contains ϵ to avoid numerical instability. A separate availability mask should be retained so that a zero weight caused by absence is distinguishable from a zero weight caused by learned unreliability.

9. Calibration-Aware Training

Calibration is the central methodological emphasis of the project. Standard anomaly-detection training optimizes predictive correctness but does not necessarily produce probabilities that correspond to empirical correctness. The proposed system therefore treats calibration as both a training consideration and a primary evaluation dimension.

9.1 Detection loss

For binary anomaly prediction, a baseline objective may be binary cross-entropy:

$$L_{\text{detection}} = \text{BCE}(y, p)$$

Focal loss may be substituted when the anomaly/normal class imbalance warrants it. The choice should be kept consistent across baselines where possible.

9.2 Calibration objective

Standard ECE is usually computed by binning predictions by confidence and comparing average confidence with empirical accuracy. Because this binning is not a straightforward differentiable operation, ordinary ECE should not simply be inserted into the computational graph and described as a backpropagated loss.

The primary proposed differentiable calibration term is the Brier score:

$$L_{\text{Brier}} = (1/N) \sum_n (p_n - y_n)^2$$

The Brier score is a proper scoring rule and can be optimized directly. ECE remains a major evaluation metric. A held-out validation set is used for post-hoc temperature scaling when appropriate, following established calibration practice.

Figure 4 – Calibration-Aware Training Objective

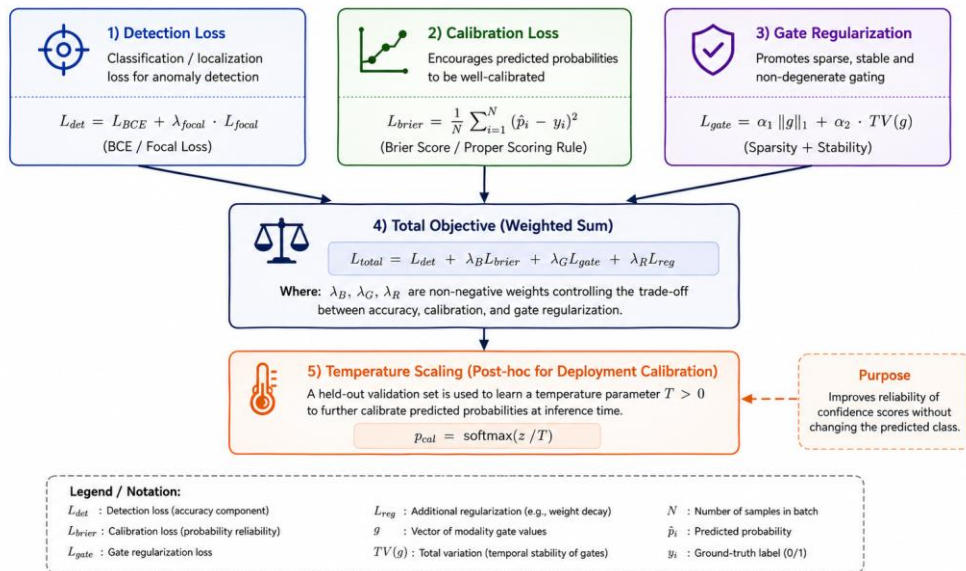


Figure 4. Calibration-aware training objective.

The design separates differentiable training losses from post-hoc calibration and evaluation. This avoids the common methodological mistake of treating ordinary binned ECE as a directly differentiable objective.

9.3 Total objective

$$L_{\text{total}} = L_{\text{detection}} + \lambda_B L_{\text{Brier}} + \lambda_{\text{gate}} L_{\text{gate}} + \lambda_{\text{reg}} L_{\text{reg}}$$

Here L_{gate} can include constraints that prevent degenerate weighting, while L_{reg} can penalize unstable or excessively sparse gate behavior. λ values must be selected on a validation set and reported transparently.

9.4 Post-hoc calibration

After model training, temperature scaling can be fit on a held-out calibration set. This is not a substitute for the calibration-aware training objective; rather, it provides a deployment-level correction mechanism. The study should report both pre- and post-temperature-scaling results so the reader can distinguish intrinsic calibration improvements from post-hoc adjustment. [\[cite?turn1search0?\]](#)

10. Handling Missing, Noisy, and Contradictory Modalities

Figure 5 — Missing / Noisy / Contradictory Modality Handling

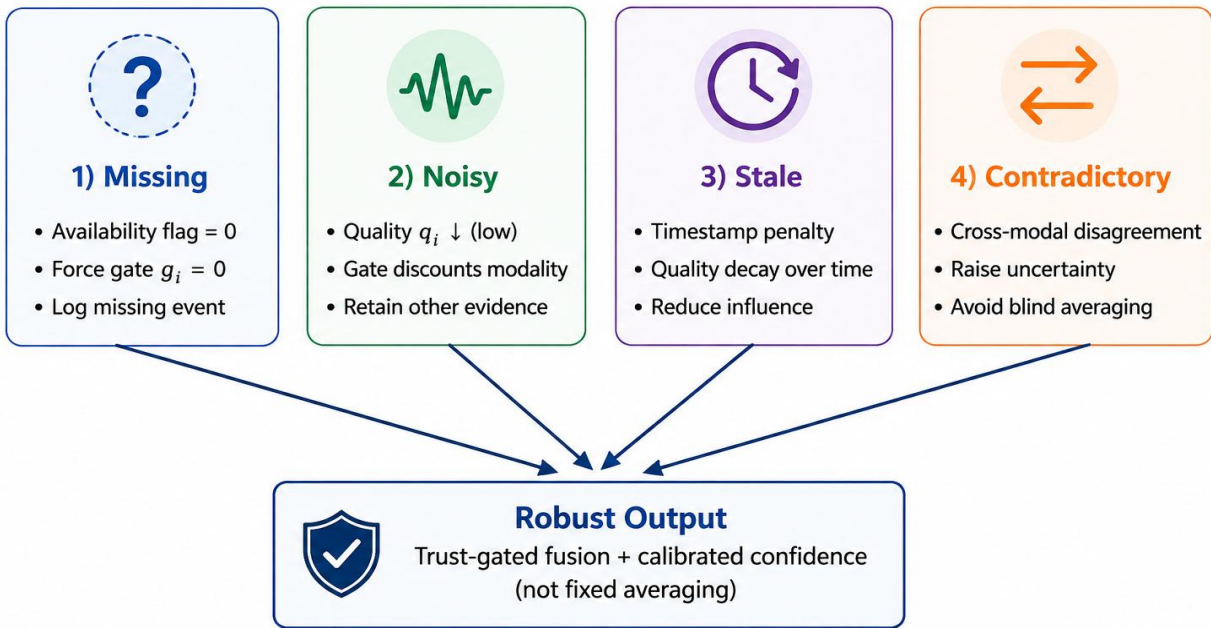


Figure 5. Missing, noisy, stale, and contradictory modality handling.

The same trust mechanism handles several degradation modes, but the logging semantics remain explicit so that later analysis can separate absence, corruption, staleness, and disagreement.

10.1 Missing modality

If a modality is completely unavailable, the system should not silently replace it with a statistically convenient imputation unless the experiment explicitly studies imputation. In the primary protocol, the availability mask is set to zero and the modality gate is forced to zero. This makes the experiment interpretable.

10.2 Noisy modality

Noise can be injected into synthetic telemetry or introduced through image corruption. The gate should respond to measurable quality changes while preserving useful complementary evidence. Noise levels should be documented and reproducible.

10.3 Stale modality

Staleness is distinct from random noise. A perfectly measured value from ten minutes ago may be less relevant than a noisy value measured now. A temporal decay feature should therefore be part of q_i . The exact decay constant should be selected according to the simulated sensor frequency and intended industrial use case.

10.4 Contradictory modalities

Consider a case in which vision indicates a surface crack while vibration remains nominal. A fixed average can reduce the anomaly score or generate an unjustifiably precise intermediate probability. The trust-gated model should examine evidence quality and context, identify disagreement, and either discount the less reliable source or produce a lower-confidence decision. The desired behavior is not necessarily to always trust vision; it is to avoid treating disagreement as if it were independent agreement.

Contradiction experiments should therefore include controlled cases where the identity of the corrupted modality is known. This allows gate behavior to be evaluated against the injected ground truth.

11. Human-in-the-Loop Trust Recalibration

The trust memory is updated only from logged deployment evidence. Each prediction record should contain timestamp, machine/plant ID, modality availability, quality features, trust weights, prediction, calibrated confidence, human correction, and final confirmed outcome where available.

Figure 6 — Human Feedback → Trust Recalibration Loop

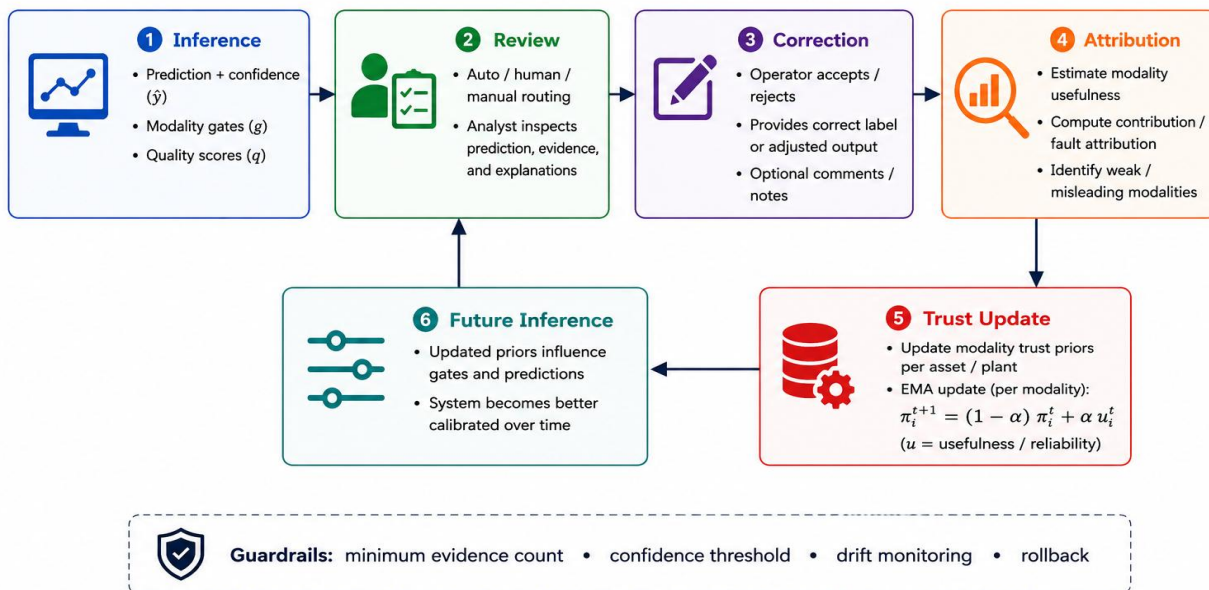


Figure 6. Human feedback to trust recalibration loop.

The global gate remains a learned model parameterization. Human feedback updates a separate asset-specific prior, making the deployment loop interpretable and easier to audit.

11.1 Primary update mechanism: guarded exponential moving average

The primary implementation should use a guarded exponential moving average because it is simple, computationally inexpensive, and easy to inspect. For modality i on asset a :

$$p_i^{\{(t+1)\}}(a) = (1-\alpha) p_i^{\{(t)\}}(a) + \alpha u_i^{\{(t)\}}(a)$$

where u_i is an estimated modality usefulness score derived from a reviewed outcome. A direct causal attribution is not available from a single correction, so u_i should be conservative. One practical definition uses the change in validation loss when the modality is removed or its weight is perturbed, combined with whether the modality's evidence agreed with the corrected outcome. This avoids pretending that a human correction uniquely identifies the causal contribution of one modality.

Bayesian updating is a viable alternative, particularly if uncertainty over the trust prior is important. Contextual bandits are another future direction if the system eventually chooses among evidence sources under explicit decision costs. For the two-person project, EMA is preferred as the primary mechanism and Bayesian/bandit approaches can be included as future work.

11.2 Global parameters versus asset priors

The global gate parameters θ are trained across the dataset and encode general reliability behavior. The asset-specific prior $p_i(a)$ is persistent state and represents local deployment experience. Conflating the two would make every human correction modify the global model and would increase the risk of catastrophic or noisy updates.

11.3 Cold-start behavior

A new machine has no trust history. The system therefore initializes priors from a plant-level or population-level prior and gradually personalizes them. The cold-start period should be explicitly identified in evaluation rather than hidden.

12. Downstream Intelligence

The downstream modules are important for platform completeness but are not the primary novelty claim.

- Known-defect detection plus open-set anomaly detection.
- Pixel-level localization/heat maps.
- Severity classification into LOW, MEDIUM, HIGH, and CRITICAL.
- Probabilistic root-cause ranking rather than causal proof.
- Failure probability over a configurable forecast horizon.
- RAG retrieval over maintenance manuals and SOPs.
- LLM-generated evidence summaries and recommended actions grounded in retrieved documents.

The RAG/LLM layer should receive structured outputs rather than raw sensor streams whenever possible. For example, the model can provide: anomaly probability = 0.87, calibrated confidence = 0.79, vision trust = 0.81, vibration trust = 0.19, severity = HIGH, candidate causes = {tool wear: 0.48, thermal drift: 0.31, contamination: 0.12}, and retrieved SOP passages. The LLM can then turn these facts into a readable explanation without becoming the detector itself.

13. Dataset and Experimental Design

Dataset / Source	Role	Strength	Limitation / Interpretation
MVTec AD	Initial vision anomaly detection	Established industrial benchmark with pixel annotations	No telemetry; cannot directly evaluate multimodal trust.
MVTec AD 2	Harder visual stress test	Challenging lighting, transparent/overlapping objects, small defects	Still image-only; no industrial telemetry.
Synthetic multimodal dataset	Primary controlled degradation benchmark	Allows paired vision/telemetry and programmable dropout/noise/staleness/conflict	Synthetic sensors are proxies, not equivalent to real factory telemetry.
Optional physical rig	Real missing/noisy modality validation	Camera + thermocouple + accelerometer on controlled object	Limited scale and ecological validity.

MVTec AD contains more than 5,000 high-resolution images across 15 object/texture categories. MVTec AD 2 provides more than 8,000 high-resolution images across eight advanced anomaly scenarios and is intended to reduce saturation of earlier benchmarks. [\[cite\]](#)[turn1search7](#)[turn2search6](#)[turn2search8](#)

The synthetic multimodal dataset is necessary because the public benchmarks do not naturally provide the desired combination of industrial images, correlated telemetry, and controlled modality degradation. Each image can be paired with simulated temperature/vibration streams whose statistical properties correlate with defect severity. The pairing must be described as a modeling assumption rather than a claim about actual industrial physics.

Programmatic corruption should include: modality dropout, Gaussian noise, burst missingness, timestamp staleness, range violations, sensor drift, and contradiction injection. The random seed, corruption distributions, and severity levels must be fixed and reported for reproducibility.

13.1 Optional physical rig

A small physical setup can include one camera, a thermocouple, an accelerometer, and a controlled object with deliberately introduced defects. Sensor feeds should be deliberately interrupted during some trials. The physical rig is not required for the core research contribution, but it substantially improves credibility by showing that the degradation protocol is not purely synthetic.

14. Experimental Evaluation

Figure 7 — Experimental Evaluation Protocol

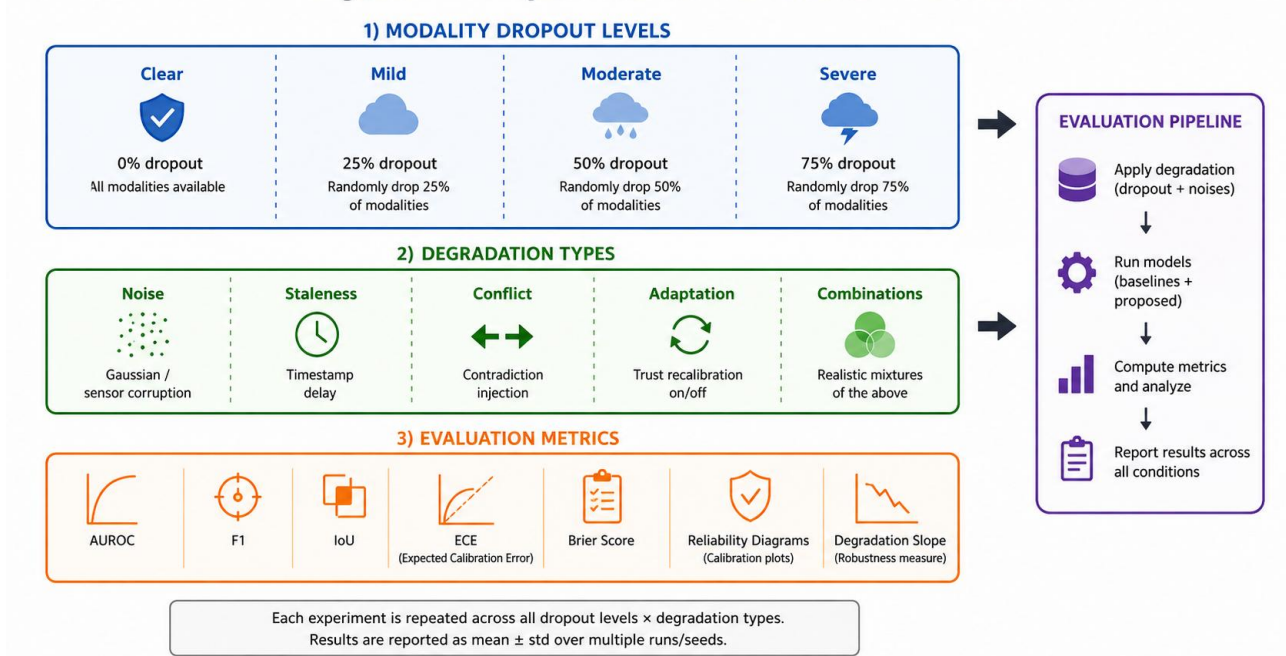


Figure 7. Experimental evaluation protocol.

The evaluation separates clean performance from degradation robustness, then adds a deployment-style trust adaptation experiment.

The central experiment should report both performance and calibration as functions of modality degradation. Every model should be evaluated under the same corruption scenarios.

Scenario	Condition	Primary question
A	0% dropout	Does the proposed model retain competitive clean-data performance?
B	25% dropout	Does dynamic trust improve over static fusion?
C	50% dropout	Does calibration degrade more slowly?
D	75% dropout	Does the model fail gracefully rather than become overconfident?
E	Contradiction	Can the gate discount unreliable evidence?
F	Noise	Does quality-aware gating suppress corrupted modalities?
G	Staleness	Does temporal quality affect trust appropriately?
H	Trust recalibration on/off	Does persistent asset trust improve behavior after repeated feedback?

The evaluation should use repeated random seeds and confidence intervals where feasible. If the project is small, at minimum report mean and standard deviation across repeated runs rather than a single best run.

15. Baselines

Baseline	Description	Purpose
B1 Image-only	Vision model without telemetry/history/documents	Measures value of multimodality.
B2 Static fusion	Fixed or equal modality weights	Tests whether dynamic weighting matters.
B3 Hybrid fusion	Reproduction-inspired hybrid RGB/3D fusion design	Connects to M3DM-style static/hybrid fusion.
B4 Crossmodal mapping	Crossmodal feature mapping baseline	Tests against cross-modal consistency.
B5 Dynamic gate, no calibration	Learned trust gate + standard detection loss	Isolates effect of calibration objective.
B6 Dynamic gate + calibration	Gate + Brier/calibration-aware training	Tests calibration contribution.
B7 Full system	Gate + calibration + asset trust + human feedback	Tests complete proposed loop.

The baseline ladder is designed so that the reader can isolate contributions rather than comparing only against an image-only model.

16. Ablation Study

Ablation	Static fusion	Dynamic gate	Calibration	Asset trust	Human feedback	What it tests
A	✓	—	—	—	—	Static multimodal reference.
B	✓	—	✓	—	—	Does calibration help without dynamic weighting?
C	—	✓	—	—	—	Effect of per-instance reliability gating.
D	—	✓	✓	—	—	Core calibration-aware gating.
E	—	✓	✓	✓	—	Effect of persistent asset-specific trust.
F	—	✓	✓	✓	✓	Full deployment adaptation loop.

The most important comparison is D versus F under increasing degradation. If the full feedback system does not improve robustness or calibration over the non-adaptive gate, the persistence mechanism should not be presented as beneficial merely because it exists.

17. Calibration Evaluation

17.1 Expected Calibration Error

$$ECE = \sum_{m=1}^M (|B_m|/N) |\text{acc}(B_m) - \text{conf}(B_m)|$$

Predictions are grouped into confidence bins B_m . ECE summarizes the difference between mean confidence and empirical correctness. Because binning choices can affect ECE, the bin count and binning method must be reported.

17.2 Brier score

$$\text{Brier} = (1/N) \sum_n (p_n - y_n)^2$$

Brier score provides a proper scoring rule and is therefore useful both as an evaluation metric and as the differentiable calibration term proposed during training.

17.3 Reliability diagrams

A reliability diagram compares predicted confidence with empirical accuracy. A perfectly calibrated system follows the diagonal. The project should produce diagrams for clean data and for each degradation level.

17.4 Calibration under dropout

The central plot is dropout percentage on the x-axis and ECE on the y-axis, comparing static fusion, dynamic gating, dynamic gating plus calibration, and the full adaptive trust system. A second plot uses dropout percentage against F1 or AUROC. The pair of plots is more informative than either alone: performance shows whether the model remains useful, while calibration shows whether its confidence remains honest.

- ECE versus modality dropout.
- Brier score versus modality dropout.
- Reliability diagrams at 0%, 50%, and 75% dropout.
- F1/AUROC versus dropout.
- Calibration gap under contradiction injection.
- Confidence distribution for correct versus incorrect predictions.

18. Graceful Degradation Slope

Graceful Degradation Slope is proposed as a custom analysis metric, not an established standard benchmark metric. Let $M(d)$ denote a performance metric such as F1 at dropout fraction d . A simple slope can be estimated from a linear fit over $d \in \{0, 0.25, 0.50, 0.75\}$:

$$\text{GDS} = \text{slope of } M(d) \text{ versus } d$$

A less negative slope indicates slower performance degradation. A corresponding calibration degradation slope can be defined for ECE:

$$\text{GDS}_{\text{ECE}} = \text{slope of } \text{ECE}(d) \text{ versus } d$$

For ECE, a smaller positive slope is preferable. The metric should be reported alongside the raw curves rather than replacing them.

19. Expected Results and Hypotheses

No experimental result is claimed at this stage. The following are hypotheses to be tested:

1. H1: Trust-gated fusion will outperform static fusion under missing and noisy modalities.
2. H2: Calibration-aware training will reduce ECE and Brier score relative to equivalent gating without the calibration term.
3. H3: The advantage of dynamic trust will increase as modality degradation increases.
4. H4: Contradictory modalities will be handled more gracefully by dynamic gating than by fixed averaging.
5. H5: Persistent asset-specific trust adaptation will improve performance and/or calibration after repeated deployment feedback.

A strong research result would be a consistent advantage across degradation conditions with statistically stable calibration improvements. A falsifying result would be that dynamic gating provides no improvement over static fusion, or that its calibration advantage disappears under controlled repeated experiments. Another possible negative result is that human-feedback trust adaptation increases instability or overfits to incorrect operator corrections. Such outcomes would be scientifically useful and must be reported rather than hidden.

20. Deployment Architecture

Figure 8 — Deployment Architecture

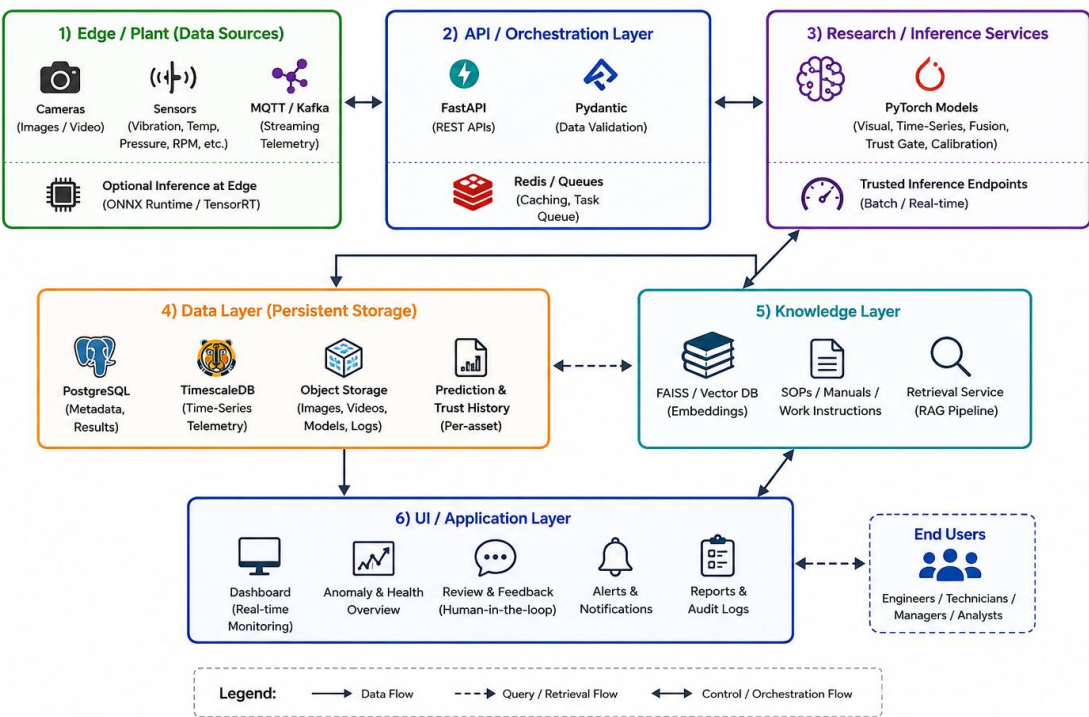


Figure 8. Deployment architecture.

The production architecture separates edge ingestion, API orchestration, research inference, persistent storage, knowledge retrieval, and the operator interface. Only the research core is mandatory for the research claim; several deployment components are optional.

Layer	Core technology	Role	Status
ML research	PyTorch, scikit-learn, NumPy/Pandas	Training and evaluation	Core
Vision	timm, torchvision, Transformers; YOLO/RT-DETR/SAM as needed	Encoding, detection, localization	Core/alternative
Anomaly baselines	PatchCore, DRAEM, PaDiM-style implementations	Reference methods	Core for comparison
Time series	PyTorch LSTM/GRU or PatchTST	Telemetry/history modeling	Core
Calibration	netcal + custom Brier/temperature scaling	ECE, Brier, reliability	Core
Backend	FastAPI, Pydantic	Inference and ingestion APIs	Engineering
Storage	PostgreSQL, TimescaleDB	Metadata and time series	Engineering
Streaming	Redis, Kafka or MQTT	Data transport	Optional/engineering
Serving	ONNX Runtime/TensorRT	Optimized inference	Optional
Frontend	React + TypeScript	Dashboard and human review	Engineering
RAG	FAISS/Chroma + sentence-transformers	SOP/manual retrieval	Engineering
LLM	Hosted or local model	Explanation layer	Engineering/optional
Tracking	MLflow or Weights & Biases	Experiments and ablations	Recommended
Deployment	Docker; Kubernetes/cloud/edge optional	Packaging and scale	Optional

The project should resist technology sprawl. The non-negotiable stack is the research core plus enough backend functionality to demonstrate it. Kubernetes, cloud orchestration, TensorRT optimization, or a large event-streaming system should not consume time before the research evaluation is complete.

21. Database and Data Flow

A conceptual relational schema should capture the relationship between assets, observations, predictions, trust states, corrections, and maintenance events.

Entity	Key fields	Purpose
Machine	machine_id, plant_id, type	Asset identity and context.
Sensor	sensor_id, machine_id, modality, calibration	Sensor metadata.
Inspection	inspection_id, machine_id, timestamp	Inspection event.
Prediction	inspection_id, anomaly_prob, calibrated_conf, severity	Model output.
ModalityQuality	inspection_id, modality, quality, availability	Evidence condition.
TrustWeight	inspection_id, modality, gate_weight, prior	Fusion state.

HumanCorrection	inspection_id, reviewer, correction, timestamp	Feedback.
MaintenanceEvent	machine_id, event_type, timestamp, outcome	Operational outcome.
Document	document_id, version, source, freshness	Knowledge source.
RetrievedEvidence	inspection_id, document_id, score, passage	RAG evidence.
FailureEvent	machine_id, timestamp, failure_type	Ground truth for predictive maintenance.

The trust loop reads Prediction, ModalityQuality, TrustWeight, HumanCorrection, and eventual FailureEvent/MaintenanceEvent records. The updated trust prior is then versioned so that the exact trust state used at any historical prediction remains auditable.

22. API Design

Endpoint	Purpose	Representative response
POST /ingest/vision	Submit image/video frame and metadata	inspection_id, image_quality
POST /ingest/telemetry	Submit timestamped sensor batch	accepted_samples, quality_flags
POST /predict	Run multimodal inference	anomaly_prob, confidence, weights, severity
GET /machine/{id}/health	Current machine state	health summary, active anomalies
GET /machine/{id}/trust	Current modality trust	prior, recent weights, trend
POST /review/correction	Submit operator review	correction_id, updated review state
GET /machine/{id}/history	Retrieve historical predictions	time series of scores and events
POST /documents/ingest	Add/update SOP/manual	document_id, version, index status
GET /inspection/{id}/explanation	Retrieve grounded explanation	evidence, recommendation, citations

The API should return calibrated confidence and modality trust separately. This prevents the frontend from accidentally treating a modality's reliability score as the probability that the anomaly prediction is correct.

23. Dashboard Design

The dashboard should make the research mechanism visible rather than displaying only a red/green machine-health indicator.

- Machine health and current operating state.
- Active anomalies with pixel-level localization.
- Calibrated anomaly confidence.
- Modality availability and quality.
- Per-modality trust weights.
- Severity and response tier.
- Probable root causes with explicit ranking semantics.
- Failure probability over the selected horizon.
- Recommended action with supporting SOP evidence.

- Historical anomaly/confidence trend.
- Human review and correction controls.
- Audit trail showing which evidence state produced each decision.

A useful review panel should show a compact evidence card: 'Vision trust 0.82 — high quality; vibration trust 0.17 — stale; history trust 0.64; document evidence 0.71. Final calibrated anomaly probability 0.86.' This is much more informative than exposing only a final confidence number.

24. Final Demonstration

6. Camera observes an industrial component.
7. Temperature/vibration sensors monitor operating conditions.
8. A controlled defect is introduced.
9. One sensor is disconnected or corrupted.
10. The model detects an anomaly.
11. The dashboard displays calibrated confidence.
12. The degraded modality is explicitly shown.
13. The trust gate reduces its influence.
14. Severity is estimated.
15. Probable causes are ranked.
16. RAG retrieves the relevant SOP.
17. The LLM generates a grounded recommendation.
18. A human reviews or corrects the prediction.
19. The correction is logged.
20. The asset-specific trust prior is updated.
21. A subsequent inference shows the updated trust state.

The demonstration should visually distinguish a static-fusion counterfactual from the proposed gate. For example, the UI can show the fixed-fusion confidence beside the trust-gated confidence under the same degraded sensor condition. The point is not to manufacture a favorable result; it is to make the hypothesis testable in front of the evaluator.

25. Limitations

- Synthetic sensor streams are controlled proxies and cannot reproduce the full physics, noise, failure modes, and temporal dependencies of factory telemetry.
- The optional physical rig provides limited validation and cannot represent the diversity of industrial equipment.
- MVTec-style images are not equivalent to deployment-specific inspection images.
- Calibration metrics depend on sample size, binning, class balance, and distribution shift.
- Quality estimators can themselves be wrong and can become a hidden failure point.
- Human correction labels can be incorrect, inconsistent, delayed, or biased.
- New machines have a cold-start trust problem.
- Distribution shift can invalidate both global gates and persistent priors.
- The fusion and quality-estimation stack increases computational and engineering complexity.
- RAG retrieval can surface incomplete or outdated procedures.

- LLMs can hallucinate even when given retrieved context; strict grounding and validation remain necessary.
- Root-cause ranking is correlational evidence and does not establish causality.

These limitations are not peripheral. They define the boundaries within which the research claim should be interpreted.

26. Ethical and Safety Considerations

Industrial AI can influence maintenance timing, production downtime, and potentially safety-critical decisions. False negatives can allow defects to escape inspection; false positives can trigger unnecessary maintenance or production interruption. The system should therefore use calibrated confidence to support human decision-making rather than present model outputs as unquestionable truth.

- Human oversight for low-confidence, high-severity, or safety-sensitive cases.
- Auditability of predictions, evidence, trust weights, and corrections.
- Protection of sensor and machine data integrity.
- Clear operator accountability boundaries.
- Versioning of models, documents, calibration parameters, and trust priors.
- Fail-safe behavior when required modalities are unavailable.
- Explicit display of uncertainty and degradation state.
- No automated maintenance recommendation should be interpreted as causal proof or a safety certification.

The LLM layer should be constrained to retrieved procedural evidence and structured model outputs. In particular, it should not invent a maintenance procedure that is absent from the approved knowledge base.

27. Two-Person Implementation Plan

Workstream	Person 1 — ML/Research	Person 2 — Systems/Product	Shared
Vision	Vision baseline, localization	Image ingestion	Dataset interface
Telemetry	Time-series encoder	Streaming ingestion	Synthetic telemetry design
Fusion	Quality gates, trust gate, fusion	Inference API integration	Architecture contracts
Calibration	Brier/ECE, temperature scaling	Expose confidence in API/UI	Evaluation
Trust	Trust-prior update and ablations	Correction logging	Review schema
RAG/LLM	Structured evidence contract	Retrieval + LLM integration	Grounding tests
Platform	Model packaging	Backend, DB, dashboard, deployment	Final integration
Research	Experiments, ablations, report	Engineering documentation	Final demo

Person 1 owns the research core and empirical validity. Person 2 owns the platform interfaces that make the research observable and deployable. Shared ownership begins at the dataset interface, because both teams depend on a stable schema for modality availability, quality, trust, prediction, correction, and outcome.

28. Development Roadmap

Phase	Technical deliverable
Phase 1 — Vision baseline	Train/evaluate image-only anomaly detection and localization on MVTec AD; establish reproducible metrics and visual outputs.
Phase 2 — Synthetic multimodal dataset	Pair visual samples with simulated telemetry; implement deterministic dropout, noise, staleness, and contradiction generators.
Phase 3 — Static fusion baseline	Implement fixed/equal-weight fusion and reproduce a representative hybrid/crossmodal baseline.
Phase 4 — Trust gates + calibration	Implement quality estimators, learned gates, Brier calibration term, held-out temperature scaling, and reliability diagrams.
Phase 5 — Degradation benchmark	Run 0/25/50/75% dropout plus noise, staleness, and contradiction experiments; freeze the evaluation protocol.
Phase 6 — Human feedback + trust recalibration	Implement correction schema, guarded EMA trust updates, asset-level priors, and simulated deployment sequences.
Phase 7 — RAG + LLM	Index SOPs/manuals and generate evidence-grounded explanations from structured outputs.
Phase 8 — Dashboard/backend	Expose trust, calibration, anomaly, severity, root-cause ranking, and review controls through APIs/UI.
Phase 9 — Optional physical rig	Collect a small real dataset with controlled defect and sensor-interruption scenarios.
Phase 10 — Final integration and report	Run final ablations, document limitations, package deployment, prepare demonstration, and write the research report/paper.

The phases are sequential in research dependency but can overlap operationally. There is deliberately no artificial short hackathon schedule. The priority is to complete the research core before expanding the product surface.

29. Contributions

Contribution	Type	Claim
Multimodal industrial anomaly architecture with explicit reliability estimation	Algorithmic / engineering	Integrates heterogeneous evidence with explicit quality state.
Per-instance trust-gated fusion	Algorithmic	Dynamically changes modality influence based on evidence and context.
Calibration-aware optimization/evaluation	Methodological	Treats calibration as a first-class objective and evaluation axis.
0/25/50/75% dropout + contradiction benchmark	Experimental	Provides a controlled degradation protocol.
Persistent asset-specific trust memory	Algorithmic / deployment	Stores machine/plant-specific modality priors.
Human correction → trust recalibration loop	Methodological / deployment	Connects deployment feedback to future fusion behavior.
End-to-end platform	Engineering	Integrates detection, localization, severity, root-cause ranking,

		predictive maintenance, RAG, review, and dashboard.
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The first two and the calibration/trust mechanisms are the strongest research claims. The RAG, LLM, dashboard, database, API, and deployment components provide engineering breadth and make the research demonstrable but should not be presented as the primary scientific novelty.

30. Conclusion

This project proposes a focused research and engineering framework for multimodal industrial anomaly intelligence under imperfect evidence. The central problem is not simply combining more modalities; it is deciding how much each modality should be trusted when its quality changes, when modalities disappear, and when evidence conflicts.

The proposed solution combines per-instance trust gating, calibration-aware optimization and evaluation, a controlled degradation benchmark, and persistent asset-specific trust updated from human corrections. Existing research establishes each of these ideas in related forms, including multimodal industrial anomaly detection, incomplete-modality learning, dynamic uncertainty-aware fusion, probabilistic multimodal fusion, and quality-aware gating. The proposed contribution is therefore deliberately framed at the level of their integration around calibrated industrial anomaly detection and a deployment feedback loop.

The research question is narrow and falsifiable: can dynamic trust reduce the performance and calibration degradation caused by missing, noisy, stale, and contradictory modalities, and can repeated machine-specific feedback improve future trust behavior? The proposed experimental design is constructed so that each part of that claim can be isolated through baselines and ablations.

A positive result would support the use of reliability-aware multimodal fusion as a practical design principle for industrial anomaly systems. A negative result would be equally informative if it demonstrates that the added complexity of trust estimation or persistent adaptation does not improve robustness or calibration. In either case, the project is structured to produce evidence rather than rely on an untestable novelty narrative.

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Bibliography verification note

The bibliography was checked against primary or high-authority sources where available, including CVF/CVPR, AAAI, Springer, Elsevier, PMLR, and arXiv. Several adjacent works supplied in the original project brief were initially described only by shorthand labels; the report resolves those labels to identifiable publications where

the match could be verified. Where a work is a preprint or adjacent-domain paper, that status is stated rather than presenting it as an established industrial benchmark.

One important correction to the initial project document is chronological: CMDIAD was accepted in 2025 but is published as an Information Fusion volume-126 article in 2026. The reference therefore uses the 2026 journal publication year. Likewise, MVTec AD 2 is now a 2026 International Journal of Computer Vision publication rather than merely a prospective benchmark.

Appendix A — Figure and Table Checklist

- Figure 1 — Overall System Architecture.
- Figure 2 — Detailed Trust-Gated Fusion Architecture.
- Figure 3 — Modality Quality Estimation Pipeline.
- Figure 4 — Calibration-Aware Training Objective.
- Figure 5 — Missing/Noisy/Contradictory Modality Handling.
- Figure 6 — Human Feedback → Trust Recalibration Loop.
- Figure 7 — Experimental Evaluation Protocol.
- Figure 8 — Deployment Architecture.
- Table 1 — Literature comparison.
- Table 2 — Proposed system modules.
- Table 3 — Dataset and data sources.
- Table 4 — Baseline comparison.
- Table 5 — Evaluation metrics.
- Table 6 — Ablation matrix.
- Table 7 — Technology stack.
- Table 8 — Risk and mitigation considerations.
- Table 9 — Two-person responsibility matrix.
- Table 10 — Research contribution matrix.

The eight figures are embedded in this report as technically oriented diagrams rather than decorative graphics. Their purpose is to make the trust/calibration research mechanism auditable to a project evaluator.

Appendix B — Risk and Mitigation Register

Risk	Mitigation
Scope creep across many platform modules	Treat trust gating, calibration, degradation evaluation, and recalibration as non-negotiable; defer dashboard/RAG/LLM polish until the research core is stable.
No physical rig	Use the controlled synthetic degradation benchmark; report the synthetic-data limitation explicitly.
Calibration implementation errors	Use established implementations for ECE/Brier where possible and validate against reference calculations.
LLM hallucination	Restrict generation to structured evidence and retrieved SOP passages; never use the LLM as the detector.

Trust prior contamination	Use confidence thresholds, repeated evidence, reviewer reliability tracking, and bounded EMA updates.
Distribution shift	Monitor calibration and trust distributions; version priors and provide rollback.
Cold start	Initialize from population/plant priors and explicitly evaluate the personalization warm-up period.

Appendix C — Required Table Completeness

Table 2. Proposed system modules

Module	Inputs	Outputs	Research/Engineering role
Visual inspection	Images/video	Defect classes + anomaly features	Core baseline
Defect localization	Image features	Pixel heatmap/mask	Core downstream
Quality estimation	Raw modality + metadata	q_V,q_S,q_H,q_D	Core research
Trust gating + fusion	z_i,q_i,asset prior	w_i,z_fused	Primary research
Calibration	Prediction logits + labels	Calibrated probability	Primary research
Severity	Anomaly + context	LOW/MEDIUM/HIGH/CRITICAL	Downstream
Root-cause ranking	Telemetry/history/anomaly evidence	Ranked causes	Downstream; correlational
Predictive maintenance	Telemetry/history	Failure probability	Downstream
RAG retrieval	Query + document store	Evidence passages	Engineering
LLM explanation	Structured outputs + evidence	Explanation/action	Engineering
Human review	Prediction + evidence	Correction/outcome	Research feedback
Trust recalibration	Correction history	Asset-specific priors	Primary research
Dashboard/API	All system outputs	Operator UI/API	Engineering

Table 5. Evaluation metrics

Metric	Definition / role	Preferred direction	Why included
AUROC	Ranking quality over anomaly scores	Higher	Standard discrimination measure
F1	Harmonic mean of precision and recall	Higher	Thresholded detection quality
Precision	Correct anomaly predictions / predicted anomalies	Higher	Controls false alarms
Recall	Correct anomaly predictions / actual anomalies	Higher	Controls missed defects
Pixel IoU	Overlap of predicted and ground-truth defect regions	Higher	Localization quality
ECE	Confidence-accuracy gap across bins	Lower	Core calibration measure
Brier score	Mean squared probabilistic error	Lower	Proper scoring rule; train + evaluate

Reliability diagram	Confidence versus empirical correctness	Closer to diagonal	Visual calibration diagnosis
Graceful Degradation Slope	Slope of performance versus dropout	Less negative	Compact robustness analysis
Calibration degradation slope	Slope of ECE versus dropout	Smaller positive	Measures confidence robustness